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Wang

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(54) **INTEGRATED INFLATING AND DEFLATING PUMP**

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F04D 25/02 (2006.01)
F04D 25/06 (2006.01)

(52) **U.S. Cl.**
CPC ***F04D 17/16*** (2013.01); ***F04D 25/02*** (2013.01); ***F04D 25/06*** (2013.01)

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,894,390	B2	11/2014	Hsu	
2004/0123396	A1 *	7/2004	Wang	F04D 25/084 5/713
2005/0079077	A1 *	4/2005	Tsai	F04D 25/084 417/423.15
2006/0204361	A1	9/2006	Xie	
2012/0304391	A1	12/2012	Driscoll, Jr.	
2020/0400150	A1 *	12/2020	Huang	F04D 29/601
2024/0309880	A1 *	9/2024	Zeng	F04D 25/068

* cited by examiner

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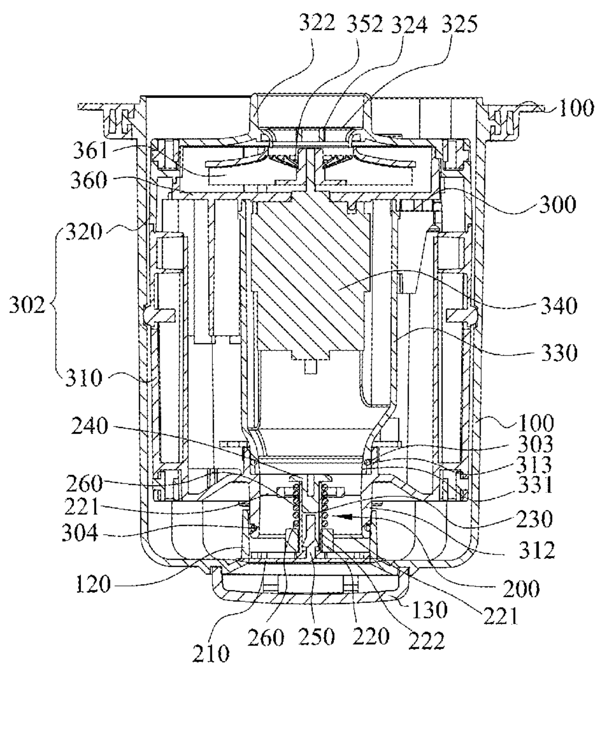
(57) **ABSTRACT**

An integrated inflating and deflating pump includes an outer cylinder, a one-way valve mechanism arranged at a bottom of the outer cylinder, and an inflating and deflating mechanism placed in the outer cylinder in a forward or reverse direction; the outer cylinder is connected to a product to be inflated or deflated; the inflating and deflating mechanism has an air inlet; when the inflating and deflating mechanism is placed in the forward direction, the air inlet is close to the product, the one-way valve mechanism seals the outer cylinder, the inflating and deflating mechanism deflates the product; when the inflating and deflating mechanism is placed in the reverse direction, the air inlet is close to the one-way valve mechanism, the one-way valve mechanism is opened, the outer cylinder is in communication with the outside, and the inflating and deflating mechanism inflates the product.

9 Claims, 10 Drawing Sheets



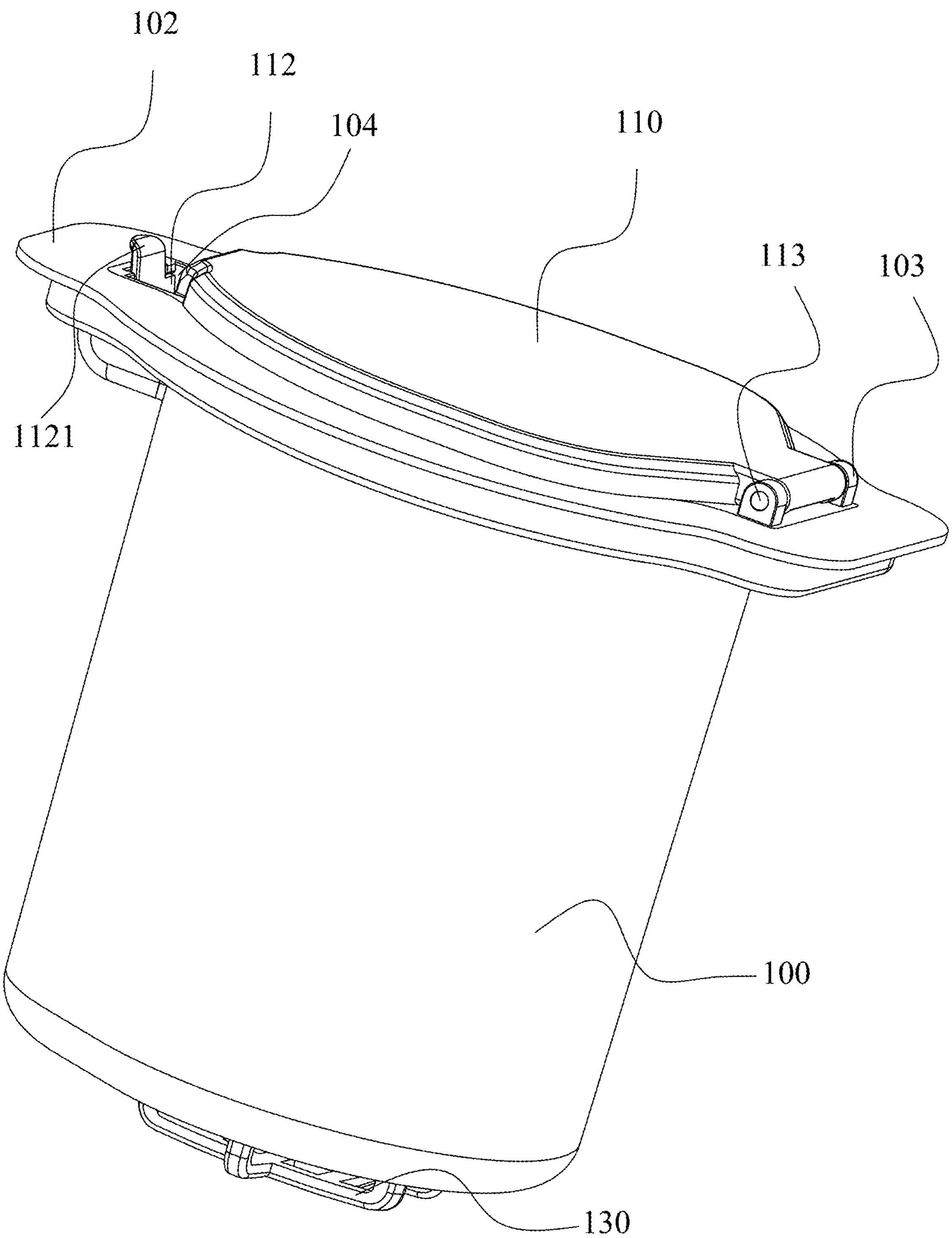


Fig. 1

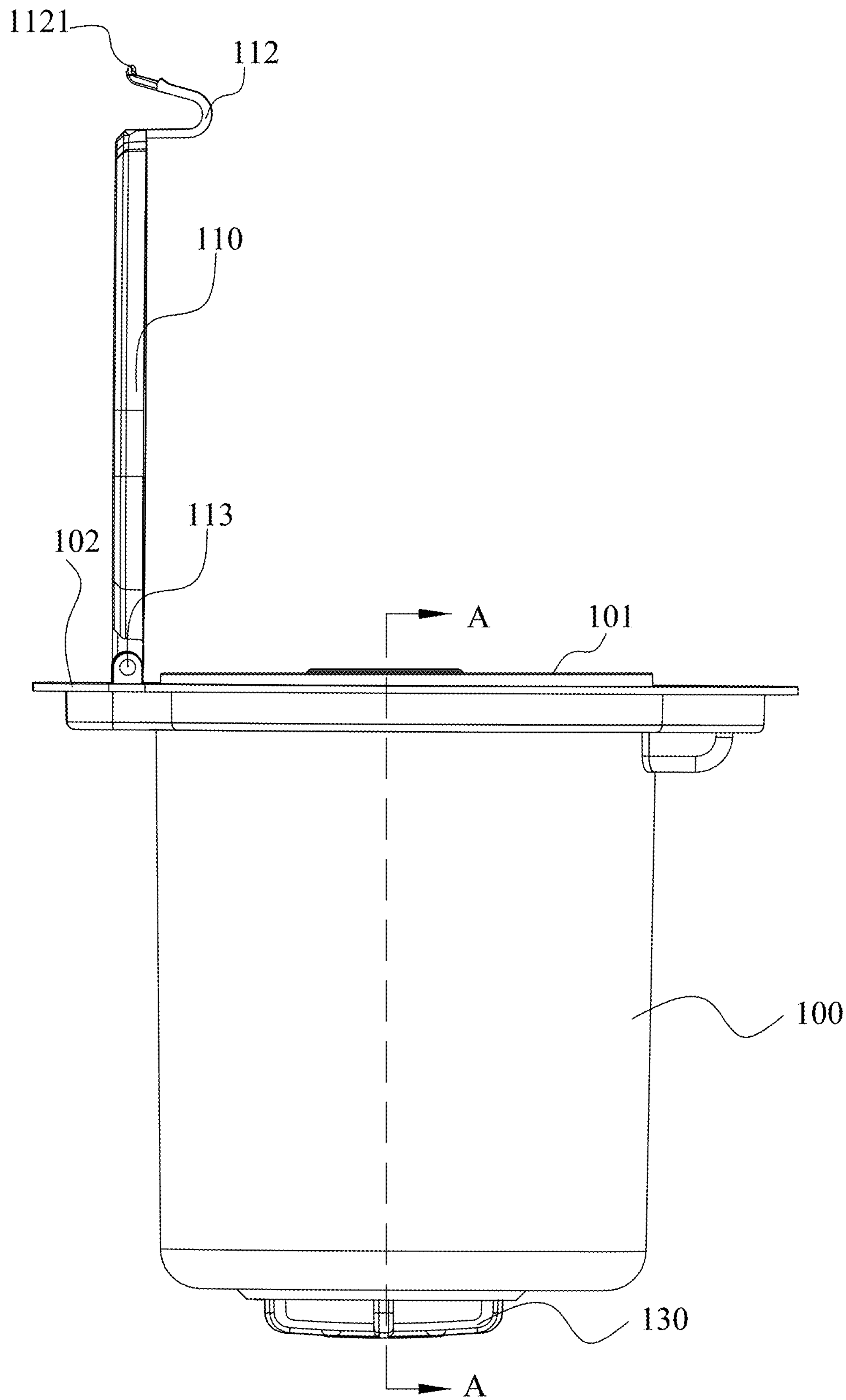


Fig. 2

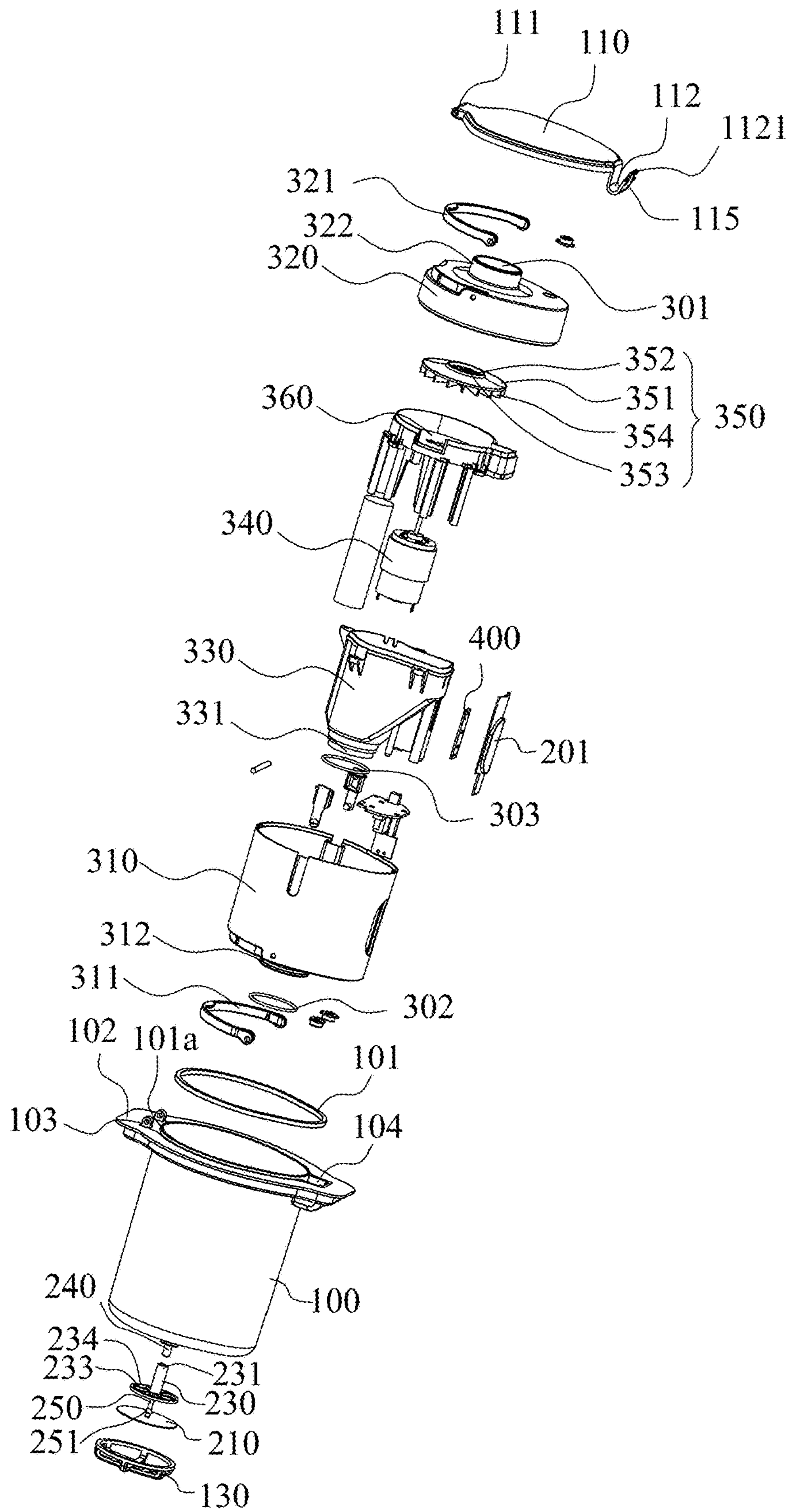

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Fig. 3

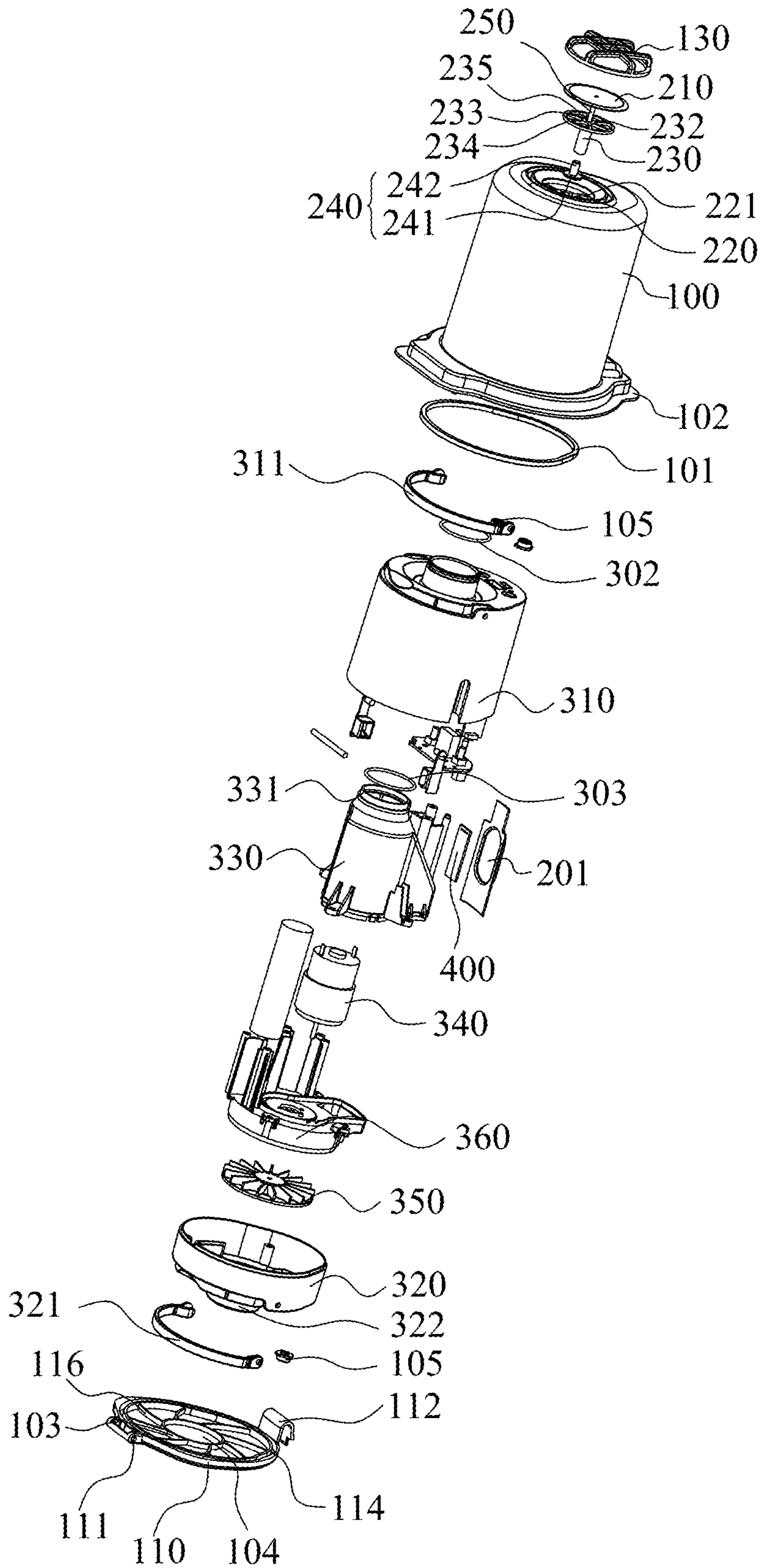
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Fig. 4

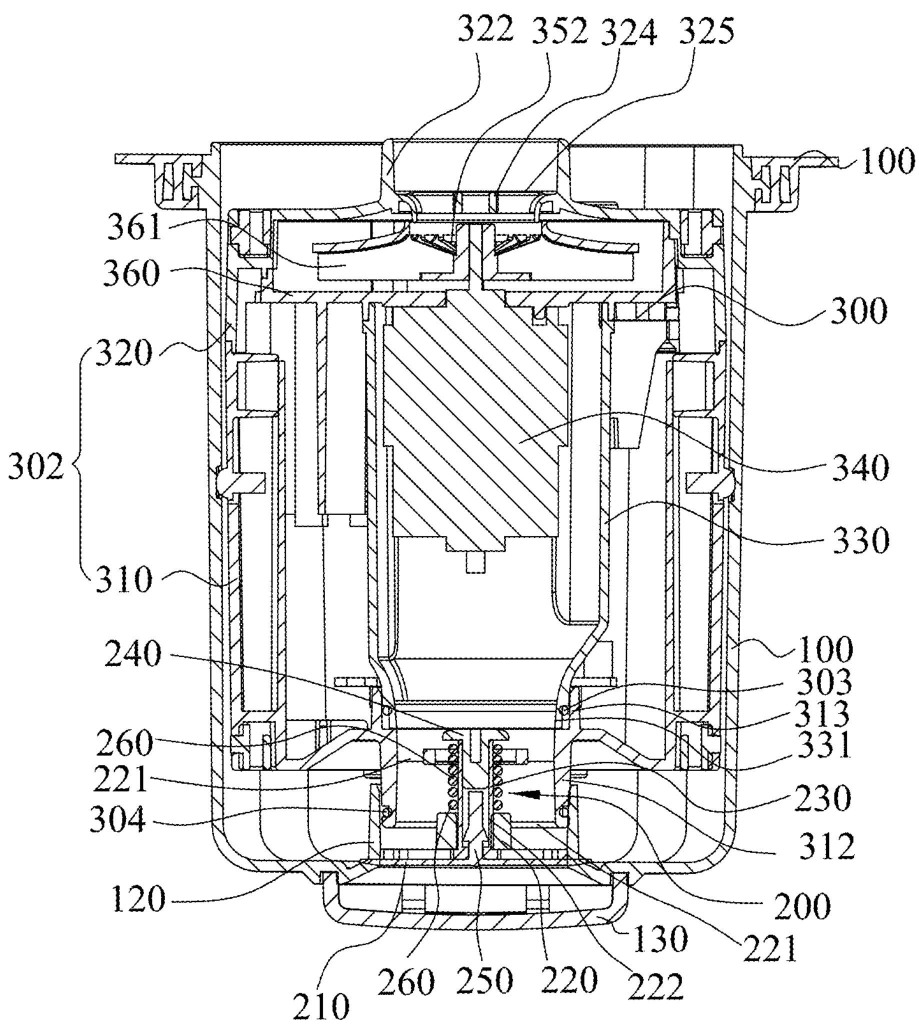
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Fig. 5

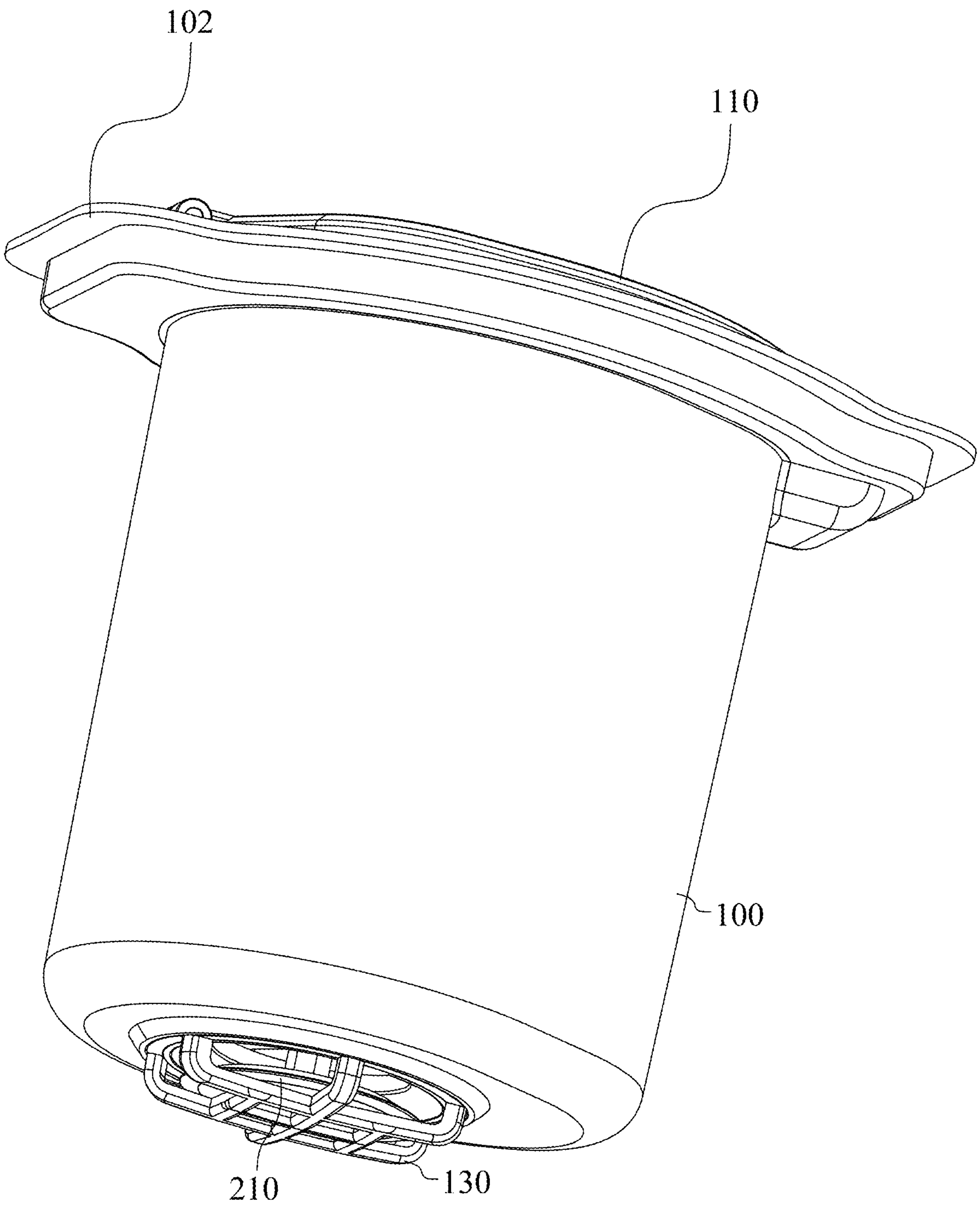

Fig. 6

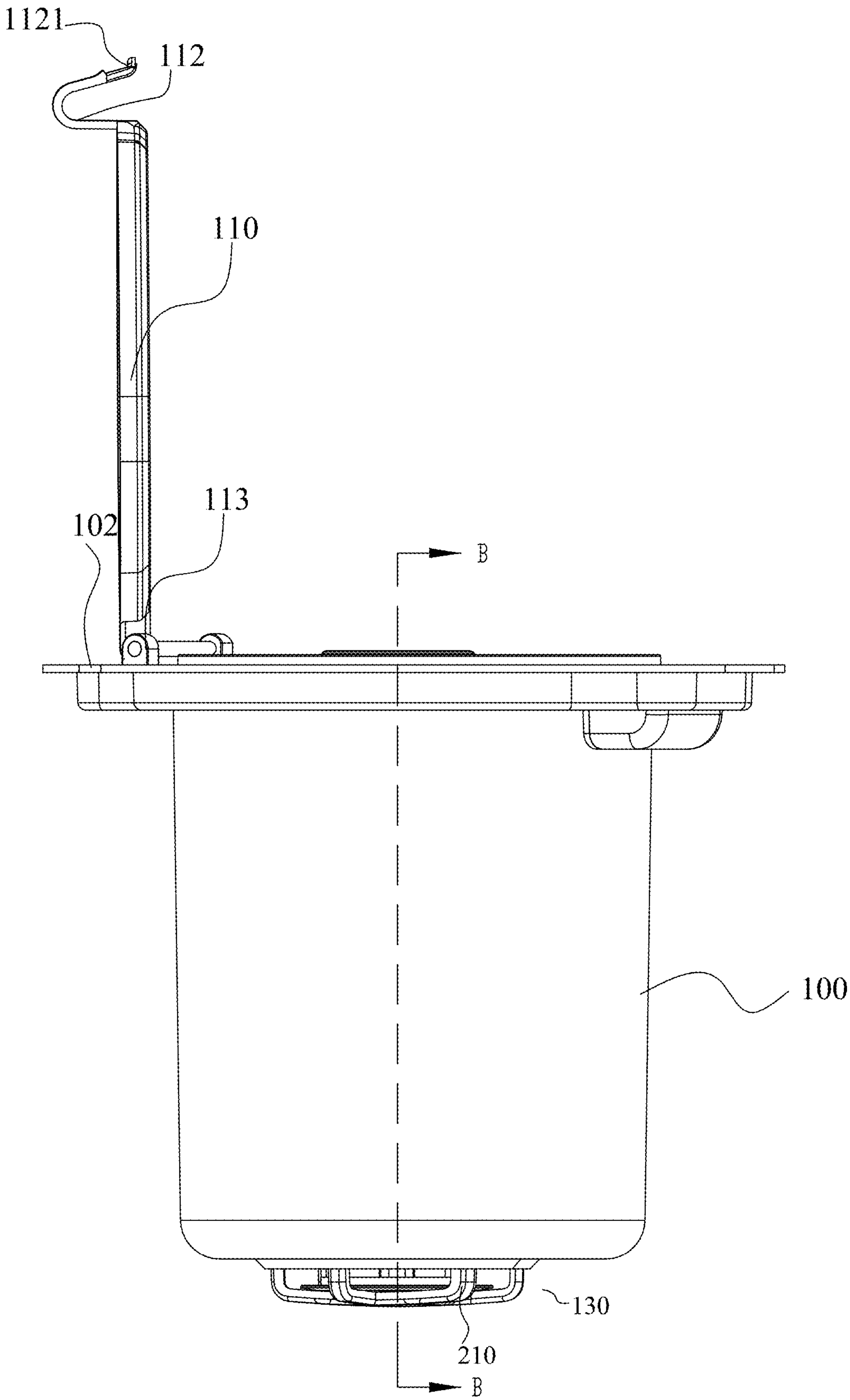

Fig. 7

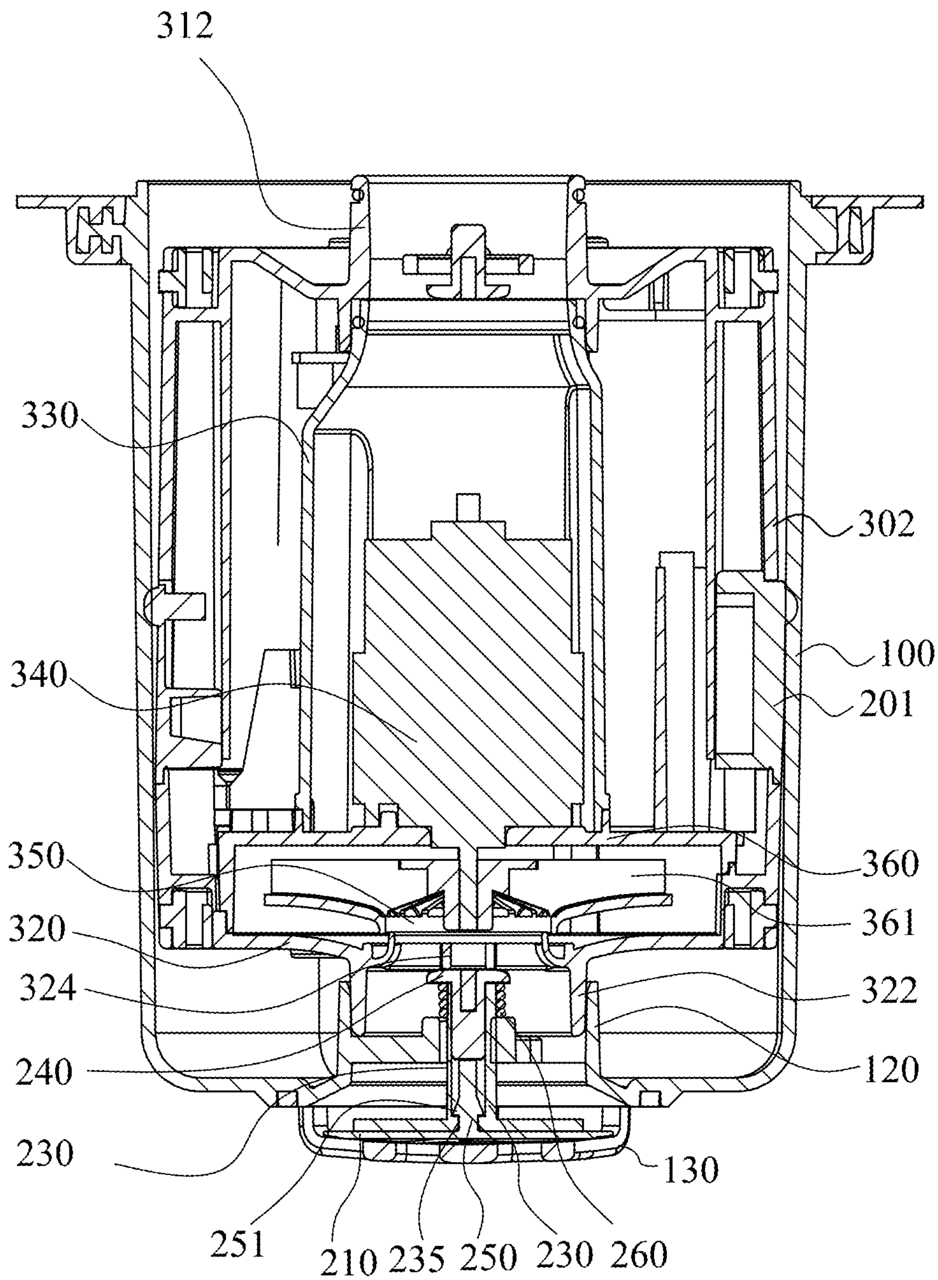
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Fig. 8

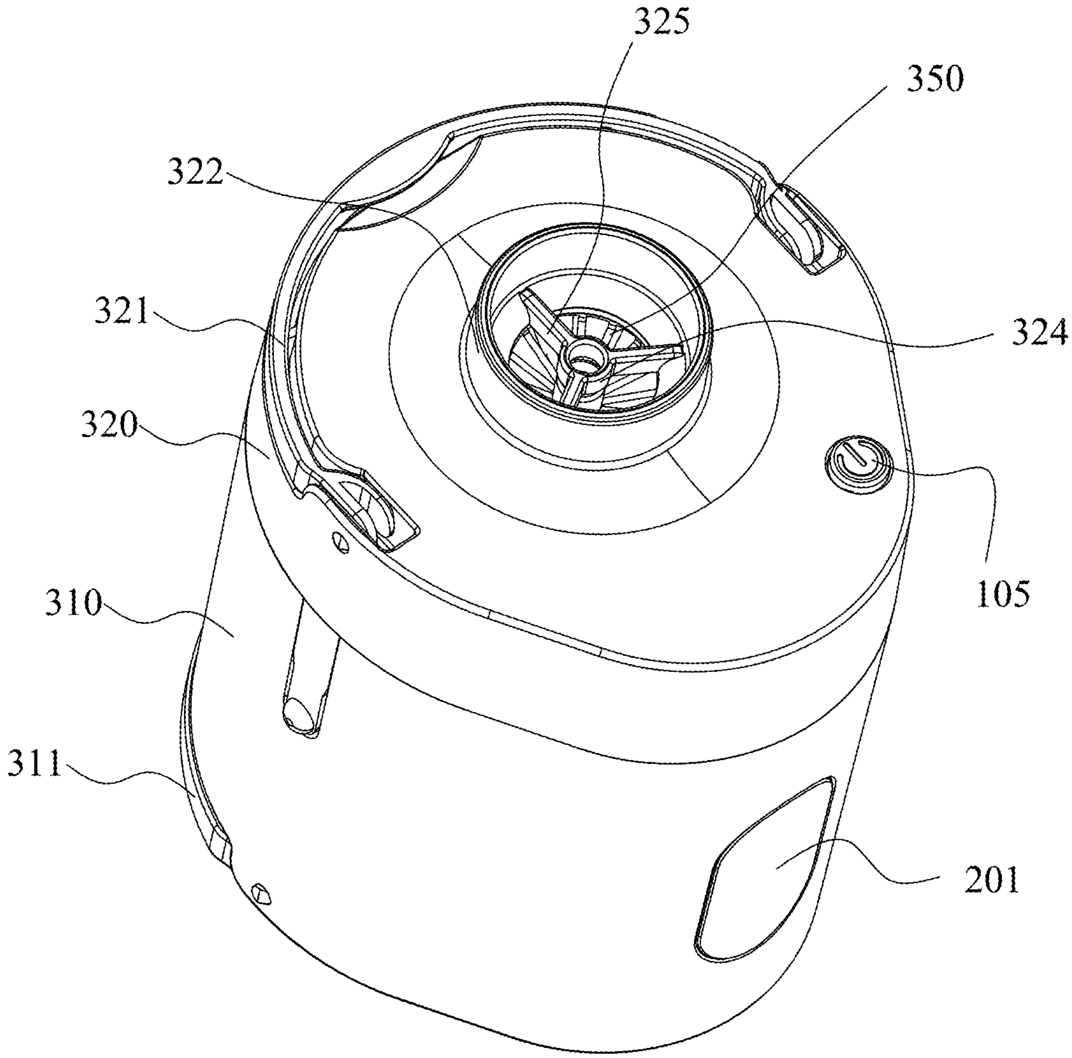

Fig. 9

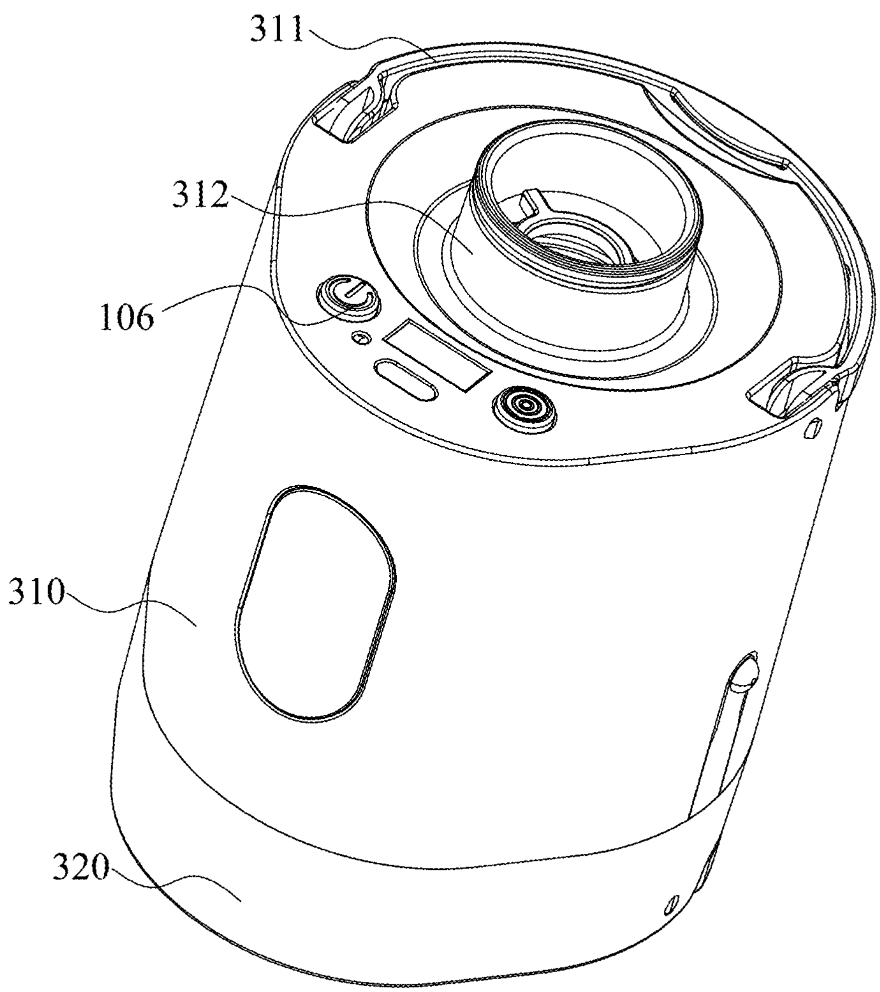

Fig. 10

INTEGRATED INFLATING AND DEFLATING PUMP

FIELD OF THE INVENTION

The present invention relates to the technical field of pumps, particularly, to an integrated inflating and deflating pump.

BACKGROUND OF THE INVENTION

An air pump is a device for inflating and deflating a closed space and includes an electric air pump, a manual air pump, a foot-driven air pump and an electric air pump. The electric air pump uses electric power to continuously compress air to generate air pressure. It is mainly used for gas spraying, sewage treatment, electroplating blast, biogas digester aeration, tunnel ventilation, etc.

Chinese Utility Model CN200820084334.1 discloses a bidirectional inflating and deflating pump, which includes a piston and an axle core. The piston is composed of a piston ring, a beam ring, a sealing sheet, and two unidirectional valve sheets. The piston ring has two openings, an air inlet chamber and an air outlet chamber are provided inside the piston, the two unidirectional valve sheets are respectively provided above and below the air inlet chamber, and two sealing sheets are respectively provided at the openings of the two unidirectional valve sheets. The axle core is provided with a partition to divide itself into two chambers, one is an air inlet channel and the other is an air outlet channel, the upper and lower ends of the air inlet channel and the air outlet channel are respectively provided with notches, the notches at the upper end are respectively in communication with an air inlet and an air outlet at a grip, and the notches at the lower ends are respectively in communication with the air inlet chamber and the air outlet chamber of the piston ring. The ventilation structure of the technical scheme in the utility model is simpler, has fewer parts, is more convenient for maintenance, and has more stable performance. However, it is necessary to provide the air inlet chamber and the air outlet chamber to achieve bi-directional inflation and deflation. The product has a large volume and a complex structure.

Chinese Utility Model CN200820145501.9 discloses an inflating and deflating pump, which includes a pump base and a pump, a partition is provided inside the pump base to divide it into an outer chamber and an inner chamber. The outer chamber is provided with a gas port in communication with the outside, and the side wall thereof is provided with a ventilation hole having a one-way valve in communication with the interior of a balloon. The air pump is arranged in the inner chamber, the partition is provided with an inlet hole and an outlet hole. The outer chamber is provided with a gas exchange valve comprising a valve seat, a valve core, and a reset spring. The valve seat is provided with an air inlet chamber and an air outlet chamber, the air inlet chamber is in communication with the inlet hole, and the wall thereof is provided with an inner airport and an outer airport, respectively in communication with the outer chamber and the airport. The air outlet chamber is in communication with the outlet port, and the wall thereof is provided with an inner airport and an outer airport, respectively in communication with the outer chamber and the airport. The valve core passes through each gas port of the valve base, and three seals are provided thereon to be opposite to each gas port. The inflating and deflating pump uses an electromagnetic valve group to control the valve core of the gas exchange

valve to convert inflation and deflation, and can control the action of the electromagnetic valve group when a motor is started, completely automated.

Most of these air pumps in the prior art are of an integrated design. The air pumps are inconvenient to be disassembled. When the air pumps are damaged, the whole devices are discarded, and the structure is complicated and the production cost is high. For this reason, it is necessary to develop an integrated inflating and deflating pump that is convenient to disassemble.

SUMMARY OF THE INVENTION

In order to overcome the deficiencies of the prior art, the present disclosure provides an integrated inflating and deflating pump to overcome the deficiencies that the pumps in the prior art are inconvenient to assemble and disassemble, using an inflating and deflating mechanism which can be placed in a forward direction and a reverse direction, the functions of inflation and deflation are converted, and the assemble and the disassemble are convenient.

The technical scheme of the present disclosure are as follows: an integrated inflating and deflating pump, which includes an outer cylinder, a one-way valve mechanism arranged at the bottom of the outer cylinder, and an inflating and deflating mechanism placed in the outer cylinder in a forward or reverse direction; the outer cylinder is connected to a product to be inflated or deflated; the inflating and deflating mechanism has an air inlet; when the inflating and deflating mechanism is placed in the forward direction, the air inlet is close to the product, the one-way valve mechanism seals the bottom of the outer cylinder, the inflating and deflating mechanism extracts air from the products to achieve deflation; when the inflating and deflating mechanism is placed in the reverse direction, the air inlet is close to the one-way valve mechanism, the one-way valve mechanism is opened, the outer cylinder is in communication with the outside, and the inflating and deflating mechanism fills the product with air from the outside to achieve inflation.

The bottom surface of the outer cylinder protrudes inwards to form a first inserted tube, the one-way valve mechanism is mounted in the first inserted tube and comprises a sealing pad, sealed and connected to the first inserted tube so as to seal the outer cylinder, or disengaged from the first inserted tube so as to open the outer cylinder.

The one-way valve mechanism further comprises a mounting seat, a valve tube, a valve cover, a valve rod, and a spring. The mounting seat is connected with the first inserted tube by several connecting ribs, the valve tube extends into the outer cylinder through the mounting seat in a sliding manner, a first end of the valve tube is plugged with the valve cover, and a second end of the valve tube is plugged with the valve rod; an end of the valve rod is provided with the sealing pad, and the valve tube is sleeved with the spring; and one end of the spring abuts against the mounting seat, and the other end thereof abuts against the valve cover.

The second end of the valve tube laterally extends several ribs to connect with a circle, when the sealing pad abuts against the ribs and the circle, it is blocked from continuing to move.

The valve rod is provided with a limiting ring, the second end of the valve tube is provided with a through hole for the valve rod to pass through, the outer diameter of the limiting ring is bigger than the inner diameter of the through hole, when the valve rod moves until the limiting ring abuts against the through hole, the limiting ring is blocked by the second end of the valve tube and cannot be detached from the through hole, thereby preventing the valve rod from falling out of the valve tube.

The inflating and deflating mechanism includes an inner cylinder comprising the first cylinder and the second cylinder that are interlocked with each other.

The bottom surface of the first cylinder is provided with a first fitting tube, the top surface of the second cylinder is provided with an air inlet tube, the opening of the air inlet tube is the air inlet, airflow enters and exits the inner cylinder through the air inlet tube.

The inflating and deflating mechanism further comprises a motor barrel, a motor and, a blade assembly. The motor barrel is accommodated in the first cylinder, the motor is mounted in the motor barrel, and a second fitting cylinder is provided at the bottom of the motor barrel; a second plug-in cylinder is provided in the first cylinder, the second fitting cylinder is inserted into the second plug-in cylinder, and a first sealing ring is provided therebetween for sealing connection; the top of the motor cylinder is provided with a top cover, and the periphery of the top cover extends towards the top surface of the second cylinder and is fixedly connected thereto so as to form a blade chamber; the blade assembly is accommodated in the blade chamber; and a rotating shaft of the motor penetrates through the top cover and used for mounting the blade assembly.

The blade assembly includes a mounting board, the center of the mounting board is provided with an airflow hole, a mounting shaft is arranged in the airflow hole. The bottom surface of the mounting board is uniformly provided with several fan blades, the blades connect with the mounting shaft, and the mounting shaft is fixed on the rotating shaft of the motor.

When the inflating and deflating mechanism is placed in the outer cylinder in the forward direction, the first fitting tube is inserted into the first inserted tube, a second sealing ring is arranged between the outer wall of the first fitting tube and the inner wall of the first inserted tube to connect them in a sealing manner. The inner wall of the lid of the outer cylinder is provided with a reinforcing ring, an upper edge of the air inlet tube is inserted in the reinforcing ring, and a gap is provided between the reinforcing ring and the upper edge of the air inlet tube so that the outer cylinder is in communication with the inner cylinder; the airflow hole directly faces the air inlet tube; and the motor rotates to drive the blade assembly to rotate, so as to extract air from the product to be deflated.

A limiting ring is provided in the air inlet tube, and the limiting ring is laterally provided with a plurality of side ribs connected to the air inlet tube. When the inflating and deflating mechanism is placed in the outer cylinder in the reverse direction, the air inlet tube is inserted into the first inserted tube, and the limiting ring pushes the valve cover, thereby pushing the valve tube away from the valve cover so that the sealing pad on the second end of the valve tube moves away from the first inserted tube, thereby opening the first inserted tube. When the motor rotates and drives the blade assembly to rotate, the air from the outside is sucked into the outer cylinder and inflated into the product to be inflated.

An air pressure display screen is arranged between the inner cylinder and the motor cylinder, and the inner cylinder is provided with a transparent window at a position directly opposite to the air pressure display screen, thereby facilitating to observe an air pressure condition in the inner cylinder.

The advantages of the present disclosure are as follows: The integrated inflating and deflating pump according to the present disclosure is provided with the inflating and deflating mechanism and the one-way valve mechanism. The inflating and deflating mechanism is installed in the outer cylinder in a forward direction or a reverse direction to convert the inflating and deflating function. The inflating and deflating mechanism is convenient to disassemble and assemble.

In the integrated inflating and deflating pump in the present disclosure, the top of the outer cylinder is connected to the product to be inflated or deflated, and the bottom of the outer cylinder is provided with the one-way valve mechanism. The one-way valve mechanism seals the bottom of the outer cylinder, and the inflating and deflating mechanism is placed in the forward direction to extract air from the product. The inflating and deflating mechanism is placed in the reverse direction, and the one-way valve mechanism is opened and is in communication with the outside so that air in the outside is sucked into the outer cylinder and inflated into the product to be inflated.

After the inflating and deflating mechanism is damaged, it is directly detached from the outer cylinder and then repaired or replaced, thereby saving resources and reducing costs.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a perspective view of an exemplary integrated inflating and deflating pump according to an embodiment of the present disclosure.

FIG. **2** is a front view of the integrated inflating and deflating pump of FIG. **1**, the lid of the integrated inflating and deflating pump is opening.

FIG. **3** is an exploded view of FIG. **1**.

FIG. **4** is the exploded view in the other direction of FIG. **1**.

FIG. **5** is a section view along the A-A line in FIG. **2**, in the view, the inflating and deflating mechanism is placed in a forward direction in the outer cylinder.

FIG. **6** is the other perspective view of the integrated inflating and deflating pump according to an embodiment of the present disclosure. This view shows the sealing pad in the one-way valve opens from the first plug-in barrel.

FIG. **7** is the front view of the FIG. **6**.

FIG. **8** is a section view along the B-B line in FIG. **6**,

FIG. **9** is a perspective view of FIG. **1** after removing the outer cylinder **100** and the lid **110**.

FIG. **10** is the perspective view in the other direction of FIG. **9**.

EMBODIMENTS

Embodiments of the present invention are described in detail below, and examples of the embodiments are shown in the attached drawings, in which the same or similar labels represent the same or similar elements or elements with the same or similar function. The embodiments described below by reference to the attached drawings are exemplary and are intended to explain embodiments of the present invention and cannot be understood as limits in the present invention.

Referring to FIGS. **1-10**, an integrated inflating and deflating pump includes an outer cylinder **100**, a lid **110** covering the top end **101***a* of the outer cylinder **100**, a one-way valve mechanism **200** arranged at the bottom of the outer cylinder **200**, and an inflating and deflating mechanism **300** placed in the outer cylinder **100** in a forward or reverse direction. The top end **101***a* of the outer cylinder **100** is connected to a product to be inflated or deflated, such as inflatable boats, swimming circles, tires, etc. The inflating and deflating mechanism **300** has an air inlet **301**; when the inflating and deflating mechanism **300** is placed in the forward direction, the air inlet **301** is close to the product to be inflated or deflated, the inflating and deflating mechanism **300** extracts air from the product to achieve deflation; when the inflating and deflating mechanism is placed in the reverse direction, the air inlet **301** is close to the one-way valve mechanism **200**, the one-way valve mechanism **200** is opened, the bottom of the outer cylinder **100** is in communication with the outside, and the inflating and deflating mechanism **300** fills the product with air from the outside to achieve inflation.

The lid **110** is pivotally connected to the top end **101***a* of the outer cylinder **100**. The lid **110** is rotated to cover and seal the outer cylinder **100**, and rotated in a reverse direction to open the outer cylinder **100**. Specifically, opposite two edges of the lids **110** are respectively provided with a shaft hole **111** and a buckle **112**. The top end **101***a* of the outer cylinder **100** extends laterally to form an ear edge **102**, the opposite two sides of the ear edge **102** are equipped with a pair of installing poles **103** and a groove **104**, a shaft **113** passes through the shaft hole **111**, with both ends thereof are respectively arranged in the installing poles **103**, thereby pivotally connecting the lid **110** to the shaft **113**. The groove **104** and the buckle **112** are matched with each other, when the lid **110** covers the outer cylinder **110**, the buckle **112** is accommodated in the groove **104** to fix the lid **110**.

The lid **110** is provided with a sealing slot **114**, the sealing slot **114** is equipped with a sealing ring **101**, when the lid **110** covers the outer cylinder **100**, the top end **101***a* of the outer cylinder **100** is squeezed into the sealing ring **101**, thereby sealing and connecting the outer cylinder **100** and the lid **110**.

As seen in FIG. **2** and FIG. **7**, when the integrated inflating and deflating pump is used to inflate or deflate the product, the lid **110** is opened from the outer cylinder **100**. The top end **101***a* of the outer cylinder **100** is connected with the product to be inflated and deflated, the sealing ring **101** also plays the function of sealing and connecting the outer cylinder **100** and the product to be inflated and deflated.

The buckle **112** is a U-shaped clip, with a free end protruding upward by a paddle **1121**, and both sides of the free end respectively protruding outward to form protrusions **115**. When the protrusions **115** enter the groove **104**, the protrusions **115** abut against the wall of the buckle **112** to achieve interference fit so that the lid **110** and the outer cylinder **100** are fixedly connected, thereby preventing detachment.

The lid **110** is provided with several reinforcing ribs **116** at one side facing the outer cylinder **100**, which increases the strength of the lid **110** and prevents excessive internal pressure in the outer cylinder **100** from blasting the lid **110**.

The bottom surface of the outer cylinder **100** protrudes inwards to form the first inserted tube **120**, and the one-way valve mechanism **200** is mounted in the first inserted tube **120**. The one-way valve mechanism **200** controls the outer cylinder **100** in communication with the outside or not.

Specifically, the one-way valve mechanism **200** comprises a sealing pad **210**, hermetically sealed the first inserted tube **120** so as to isolate the inner cavity of the outer cylinder **100** from the outside, or open the first inserted tube **120** so as to communicate the inner cavity of the outer cylinder **100** with the outside. When deflating the product, the inflating and deflating mechanism **300** extracts the air from the product, opens the sealing pad **210** by the wind pressure generated by the following blade assembly, thereby communicating the outer cylinder with the outside. When inflating the product, the inflating and deflating mechanism **300** opens the sealing pad **210**, thereby communicating the outer cylinder with the outside.

The one-way valve mechanism **200** includes a mounting seat **220**, a valve tube **230**, a valve cover **240**, a valve rod **250**, and a spring **260**. The mounting seat **200** is connected with the first inserted tube **120** by several connecting ribs **221**, thereby fixing the mounting seat **220** in the first inserted tube **120**. This hollow structural design facilitates the air in and out of the outer cylinder **100** on one hand and is used for installing the valve tube **230** described below on the other hand.

The mounting seat **220** is provided with a through hole **222**, the valve tube **230** is mounted in the through hole **222**. The first end **231** of the valve tube **230** is plugged with the valve cover **240**. The valve cover **240** includes a cover body **241** and a rod **242**, the cover **241** is located outside the first end **231** of the valve tube **230**, and the rod **242** is inserted into the first end **231** of the valve tube **230**. The second end **232** of the valve tube **230** is plugged with the valve rod **250**. The end of the valve rod **250** is provided with the sealing pad **210**, and the valve tube **230** is sleeved with the spring **260**. One end of the spring **260** abuts against the mounting seat **220**, and the other end thereof abuts against the cover body **241**.

Referring to FIG. **5**, when the integrated inflating and deflating pump is used to deflate the product, the inflating and deflating mechanism **300** is placed in the forward direction in the outer cylinder **100**, the spring is in a naturally or almost naturally straight state. The cover **240** is pushed to move away from the sealing pad **210** by the spring **260**, and the sealing pad **210** hermetically seals the first inserted tube **120**. Preferably, the second end **232** of the valve tube **230** laterally extends several ribs **233** connecting with a circle **234**. When the sealing pad **210** abuts against the ribs **233** and the circle **234**, it is blocked by the ribs **233** and the circle **234**, thereby defining the location of the sealing pad **210** in the first inserted tube **120**.

Referring to FIG. **8**, when the integrated inflating and deflating pump is used to inflate the product, the inflating and deflating mechanism **300** is placed in the reverse direction in the outer cylinder **100**, and the inflating and deflating mechanism **300** pushes the valve cover **240** of the one-way valve mechanism **200** towards the bottom of the outer cylinder **100**, the spring **260** is compressed, and the valve cover **240** pushes the valve rod **250** towards the bottom of the outer cylinder **100**, and the valve rod **250** drives the sealing pad **210** to detach from the first inserted tube **120** so that the outer cylinder **110** is in communication with the outside.

The valve rod **250** is provided with a limiting ring **251**, the second end **232** of the valve tube **230** is provided with a through hole **235** for the valve rod **250** to pass through, the outer diameter of the limiting ring **251** is bigger than the inner diameter of the through hole **235**, when the valve rod **251** moves until the limiting ring **251** abuts against the through hole **235**, the limiting ring **251** is blocked by the second end **232** and cannot be detached from the through hole **235**, thereby preventing the valve rod **250** from falling out of the valve tube **230**.

Preferably, a protective barrier **130** is fixed on the bottom surface of the outer cylinder **100**, which is located outside the sealing pad **210** to protect the sealing pad **210** and the valve rod **250**, preventing the valve rod **250** from being quickly pushed away from the first inserted tube **120**.

After the inflation is finished, the inflating and deflating mechanism **300** is disassembled, the spring **260** pushes the valve cover **240** to reset by its elastic potential energy.

Referring to FIGS. **3-5** and **8-10**, the inflating and deflating mechanism **300** includes an inner cylinder **302** comprising the first cylinder **310** and the second cylinder **320** that are interlocked with each other. The first cylinder **310** and the second cylinder **320** are respectively equipped with a first handle **311** and a second handle **321**, which are convenient for lifting the first cylinder **310** and the second cylinder **320** for installation.

The bottom surface of the first cylinder **310** is provided with a first fitting tube **312**, the top surface of the second cylinder **320** is provided with an air inlet tube **322**, the opening of the air inlet tube **322** is the air inlet **301**, the air enters and exits the inner cylinder **302** through the air inlet tube **322**.

The inflating and deflating mechanism **300** further comprises a motor barrel **330**, a motor **340**, and a blade assembly **350**. The motor barrel **330** is accommodated in the first cylinder **310**, and a second fitting cylinder **331** is provided at the bottom of the motor barrel **330**. A second plug-in cylinder **313** is provided in the first cylinder **310**. The second fitting cylinder **331** is inserted into the second plug-in cylinder **313**, and a first sealing ring **303** is provided therebetween for sealing and connecting them. The top of the motor barrel **330** is provided with a top cover **360**, and the periphery of the top cover **360** extends towards the top surface of the second cylinder **320** and is fixedly connected thereto so as to form a blade chamber **361**; the blade assembly **350** is accommodated in the blade chamber **361**.

The motor **340** is mounted in the motor barrel **330**, and a rotating shaft **113** of the motor **340** penetrates through the top cover **360** to mount the blade assembly **350**. The blade assembly **350** includes a mounting board **351**, the center of the mounting board **351** is provided with an airflow hole **352**, and a mounting shaft **353** is arranged in the airflow hole **352**. The bottom surface of the mounting board **351** is uniformly provided with several fan blades **354**, the blades **354** connect with the mounting shaft **353**, the mounting shaft **353** is fixed on the rotating shaft **113** of the motor **340**.

When the inflating and deflating mechanism **300** is placed in the outer cylinder **100** in the forward direction, the first cylinder **310** closes to the one-way valve mechanism **200**, the second cylinder **320** closes to the lid **110**, the first fitting tube **312** inserts into the first inserted tube **120**. A second sealing ring **304** is arranged between the outer wall of the first fitting tube **312** and the inner wall of the first inserted tube **120** to connect them in a sealing manner. The top end **101***a* of the outer cylinder **100** is connected to the product to be inflated or deflated; the airflow hole **352** directly faces the air inlet tube **322**; and the motor **340** rotates to drive the blade assembly **350** to rotate, so as to extract the air from the product and open the sealing pad **210** by wind pressure to achieve deflation.

A limiting ring **324** is provided in the air inlet tube **322**, and the limiting ring **324** is laterally provided with a plurality of side ribs **325** connected to the air inlet tube **322**.

Referring to FIG. **8**, when the inflating and deflating mechanism **300** is placed in the outer cylinder **100** in the reverse direction, the air inlet tube **322** is inserted into the first inserted tube **120**, and the limiting ring **324** pushes the valve cover **240**, thereby pushing the valve tube **230** to move away from the valve cover **240** so that the sealing pad **210** on the second end of the valve tube **230** move away from the first inserted tube **120**, thereby opening the first inserted tube **120**. When the motor **340** rotates and drives the blade assembly **350** to rotate, the air from the outside is sucked into the outer cylinder **100** and inflated the product to be inflated.

An air pressure display screen **400** is arranged between the inner cylinder **302** and the motor barrel **330**, and the inner cylinder **302** is provided with a transparent window **201** at a position directly opposite to the air pressure display screen **400**, thereby facilitating to observe an air pressure condition in the inner cylinder **30**.

The first cylinder **310** and the second cylinder **320** are respectively equipped with switch buttons **105** and **106**, which are respectively electrically connected to the motor **340** to control the start and stop of the motor **340**, that is, to control the inflating and deflating mechanism **300**.

A battery is placed in the integrated inflating and deflating pump and electrically connected to the motor.

A PCB board is placed in the integrated inflating and deflating pump and electrically connected to the motor.

The above content is a further detailed explanation of the present invention in conjunction with specific preferred embodiments, and it cannot be considered that the specific implementation of the present invention is limited to these explanations. For ordinary technical personnel in the technical field to which the present invention belongs, without departing from the concept of the present invention, its architectural form can be flexible and varied, and a series of products can be derived. Just making a few simple deductions or substitutions should be considered as belonging to the scope of patent protection of the present invention determined by the submitted claims.

The invention claimed is:

1. An integrated inflating and deflating pump, comprising:

an outer cylinder;

a one-way valve mechanism, arranged at a bottom of said outer cylinder; and,

an inflating and deflating mechanism, placed in said outer cylinder in a forward or reverse direction;

said outer cylinder is connected to a product to be inflated or deflated; said inflating and deflating mechanism has an air inlet; when said inflating and deflating mechanism is placed in said forward direction, said air inlet is close to said product, said inflating and deflating mechanism extracts air from said products to achieve deflation; when said inflating and deflating mechanism is placed in said reverse direction, said air inlet is close to said one-way valve mechanism, said one-way valve mechanism is opened, said outer cylinder is in communication with an outside, and said inflating and deflating mechanism fills said product with air from the outside to achieve inflation;

said bottom surface of said outer cylinder protrudes inwards to form a first inserted tube, said one-way valve mechanism is mounted in said first inserted tube and comprises a sealing pad, hermetically sealed said first inserted tube so as to isolate an inner cavity of said outer cylinder from said outside, or open said first inserted tube so as to communicate said inner cavity of said outer cylinder with said outside.

2. The integrated inflating and deflating pump according to claim **1**, wherein said one-way valve mechanism further comprises a mounting seat, a valve tube with a first end and a second end, a valve cover, a valve rod, and a spring; said mounting seat is connected with said first inserted tube by several connecting ribs, said valve tube extends into said outer cylinder through said mounting seat in a sliding manner, said first end thereof is plugged with said valve cover, and said second end thereof is plugged with said valve rod; an end of said valve rod is provided with said sealing pad, and said valve tube is sleeved with said spring; one end of said spring abuts against said mounting seat and the other end thereof abuts against said valve cover.

3. The integrated inflating and deflating pump according to claim **2**, wherein said second end of said valve tube laterally extends several ribs to connect with a circle, when said sealing pad abuts against said ribs and said circle, said sealing pad is blocked.

4. The integrated inflating and deflating pump according to claim **2**, wherein said valve rod is provided with a limiting ring, said second end of said valve tube is provided with a through hole for said valve rod to pass through, an outer diameter of said limiting ring is bigger than an inner diameter of said through hole, when said valve rod moves until said limiting ring abuts against said through hole, said limiting ring is blocked by said second end of said valve tube and cannot be detached from said through hole, thereby preventing said valve rod from falling out of said valve tube.

5. The integrated inflating and deflating pump according to claim **1**, wherein said inflating and deflating mechanism includes an inner cylinder comprising a first cylinder and a second cylinder that are interlocked with each other;

a bottom surface of said first cylinder is provided with a first fitting tube, a top surface of said second cylinder is provided with an air inlet tube, an opening of said air inlet tube is said air inlet, said air enters and exits said inner cylinder through said air inlet tube;

said inflating and deflating mechanism further comprises a motor barrel, a motor, and a blade assembly; said motor barrel is accommodated in said first cylinder, said motor is mounted in said motor barrel, and a second fitting cylinder is provided at said bottom of said motor barrel;

a second plug-in cylinder is provided in said first cylinder, said second fitting cylinder is inserted into said second plug-in cylinder, and a first sealing ring is provided therebetween for sealing and connecting said second fitting cylinder and said second plug-in cylinder; a top of said motor barrel is provided with a top cover, and a periphery of said top cover extends towards a top surface of said second cylinder and is fixedly connected thereto so as to form a blade chamber; said blade assembly is accommodated in said blade chamber; and a rotating shaft of said motor penetrates through said top cover and used for mounting said blade assembly.

6. The integrated inflating and deflating pump according to claim **5**, wherein said blade assembly includes a mounting board, a center of said mounting board is provided with an airflow hole, a mounting shaft is arranged in said airflow hole; a bottom surface of said mounting board is uniformly provided with several blades, said blades connect with said mounting shaft, said mounting shaft is fixed on said rotating shaft of said motor.

7. The integrated inflating and deflating pump according to claim **6**, wherein when said inflating and deflating mechanism is placed in said outer cylinder in said forward direction, said first fitting tube is inserted into said first inserted tube, a second sealing ring is arranged between an outer wall of said first fitting tube and an inner wall of said first inserted tube to connect them in a sealing manner; said airflow hole directly faces said air inlet tube; and said motor rotates to drive said blade assembly to rotate so as to extract said air from said product.

8. The integrated inflating and deflating pump according to claim **5**, wherein a limiting ring is provided in said air inlet tube, and said limiting ring is laterally provided with a plurality of side ribs connected to said air inlet tube; when said inflating and deflating mechanism is placed in said outer cylinder in said reverse direction, said air inlet tube is inserted into said first inserted tube, and said limiting ring pushes said valve cover, thereby pushing said valve tube away from said valve cover so that said sealing pad on said second end of said valve tube moves away from said first inserted tube, thereby opening said first inserted tube; when said motor rotates and drives said blade assembly to rotate so as to suck said air from said outside into said outer cylinder and inflate said product.

9. The integrated inflating and deflating pump according to claim **5**, wherein an air pressure display screen is arranged between said inner cylinder and said motor barrel, and said inner cylinder is provided with a transparent window at a position directly opposite to said air pressure display screen, thereby facilitating to observe an air pressure condition in said inner cylinder.

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